

Unit V

SEARCHING, SORTING AND HASHING TECHNIQUES

Course objective: To understand the concepts of search, sort and hashing

Course Outcome: Implement in searching, sorting operations and hash tables

Bloom's Taxonomy Level: K1, K2 and K3

PART A

Q.NO	Question	BTL				
1.	Give the types of sorting. Differentiate it.	K1				
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2.	What is meant by sorting and searching? Sorting: Sorting is a technique for arranging data in particular order. Order means the arrangement of data. The sorting order can be ascending or descending. The ascending order means arranging the data in increasing order whereas descending order means arranging the data in decreasing order. Searching: Searching is the technique for finding particular element from the set.	K1				

3.	What are the types of searching? Give its difference. Types: a. Linear Search b. Binary Search <table border="1"> <thead> <tr> <th>Sr.No.</th><th>Linear Search</th><th>Binary Search</th></tr> </thead> <tbody> <tr> <td>1.</td><td>For searching the element by using linear search method it is not required to arrange the elements in some specific order</td><td>The elements need to be arranged either in ascending or descending order.</td></tr> <tr> <td>2.</td><td>Each and every element is compared with the key element from the beginning of the list.</td><td>The list is subdivided into two sublists. The key element is searched in the sublist.</td></tr> <tr> <td>3.</td><td>Less efficient method.</td><td>Efficient method.</td></tr> <tr> <td>4.</td><td>Simple to implement.</td><td>Additional computation is required for computing mid element.</td></tr> </tbody> </table>	Sr.No.	Linear Search	Binary Search	1.	For searching the element by using linear search method it is not required to arrange the elements in some specific order	The elements need to be arranged either in ascending or descending order.	2.	Each and every element is compared with the key element from the beginning of the list.	The list is subdivided into two sublists. The key element is searched in the sublist.	3.	Less efficient method.	Efficient method.	4.	Simple to implement.	Additional computation is required for computing mid element.	K1
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4.	What are the types of sorting available? ● Insertion sort ● Merge sort ● Quick sort ● Radix sort ● Heap sort ● Selection sort ● Bubble sort.	K1															
5.	Define bubble sort. ● Bubble sort is the one of the easiest sorting method. In this method each fate item is compared with its neighbor and if it is a descending order sorting, then the bigger element is moved to the top of all. ● The smaller numbers are slowly moved to the bottom position. Hence it is also called as the exchange sort.	K1															
6.	Mention one advantage and disadvantage of using quadratic probing.	K1															

	Advantage: The problem of primary clustering is eliminated. Disadvantage: There is no guarantee of finding an unoccupied cell once the table is nearly half full	
7.	Define insertion sort. Give its advantages. • Successive element in the array to be sorted and inserted into its proper place with respect to the other already sorted element. • We start with the second element and put it in its correct place, so that the first and second elements of the array are in order. Advantages: • Simple to implement • This method is efficient when we want to sort small number of elements and this method has excellent performance on almost sorted list of elements.	K1
8.	Define selection sort. Scan the array to find its smallest element and swap it with the first element. Then, starting with the second element scan the entire list to find the smallest element and swap it with the second element. Then starting from the third element the entire list is scanned in order to find the next smallest element. Continuing in this fashion we can sort the entire list.	K1
9.	What do you mean by quadratic probing? Quadratic probing is an open addressing collision resolution strategy in which $F(i) = i^2$. There is no guarantee of finding an empty cell once the table gets half full if the table size is not prime. This is because at most half of the table can be used as alternative locations to resolve collisions	K1
10.	Explain why binary search cannot be performed using linked list. In binary search algorithm, the mid element needs to be searched. If binary search is implemented using arrays then by simply saying a [mid] we can access the middle element of an array in constant time. But for finding the mid element of a linked list we have to execute separate algorithm and it cannot be done in constant time. Thus implementing binary search using	K2

	linked list is very inefficient way. Hence it is not preferred to implement a binary search using linked list.	
11.	What is overflow in hashing? The situation in which there is no room for a new pair in the hash table is called overflow.	K1
12.	What is collision in hashing? The situation in which the hash function returns the same hash key for more than one record is called collision.	K1
13.	What is hashing? Hashing is a technique of storing the elements directly at the specific location in the hash table. The hashing makes use of hash function to place the record at its position. Using the same hash function the data can be retrieved directly from the hash table.	K1
14.	Define hash table. Hash table is a data structure used for storing and retrieving data very quickly. Insertion of data in the hash table is based on the key value. Hence every entry in the hash table is associated with some key. For example for storing an employee record in the hash table the employee ID will work as a key.	K1
15.	How the insertion sort is done with the array? It sorts a list of elements by inserting each successive element in the previously sorted sublist. Consider an array to be sorted $A[1], A[2], \dots, A[n]$ • Pass 1: $A[2]$ is compared with $A[1]$ and placed them in sorted order. • Pass 2: $A[3]$ is compared with both $A[1]$ and $A[2]$ and inserted at an appropriate place. This makes $A[1], A[2],$ and $A[3]$ as a sorted sub array.	K1

	• Pass n-1: A[n] is compared with each element in the sub array A [1], A [2] ...A [n-1] and inserted at an appropriate position.	
16.	What is open addressing? Open addressing is also called closed hashing, which is an alternative to resolve the collisions with linked lists. In this hashing system, if a collision occurs, alternative cells are tried until an empty cell is found.	K1
17.	Define Searching. Searching for data is one of the fundamental fields of computing. Often, the difference between a fast program and a slow one is the use of a good algorithm for the data set. Naturally, the use of a hash table or binary search tree will result in more efficient searching, but more often than not an array or linked list will be used. It is necessary to understand good ways of searching data structures not designed to support efficient search.	K1
18.	What is rehashing? Give its advantages. Rehashing is a technique in which the table is resized. That means the size of the table is doubled by creating a new table. The total size of the table is usually a prime number. Following are the situations in which the rehashing is required- • When table is completely full. • With quadratic probing the table is filled half. When insertions fail due to overflow.	K1
19.	What are the advantages and Disadvantages of Separate Chaining and linear probing • When using linear probing, one advantage is that it can use less memory than separate chaining. When using separate chaining, each index of the hash map stores another data structure, which can require a lot more memory than linear probing does, which only needs a single list data structure. • Linear probing perform faster than separate chaining.	K1

20.	State the complexity of binary search. Binary search runs in at worst logarithmic time, making $O(\log n)$ comparisons, where n is the number of elements in the array, the O is Big O notation, and $\log$ is the logarithm. Binary search takes constant ($O(1)$) space, meaning that the space taken by the algorithm is the same for any number of elements in the array.	K2
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PART B

Q.NO	Question	BTL
1.	Explain in detail about sorting and different types of sorting techniques	K2
2.	Demonstrate the insertion sort results for each insertion for the following initial array of elements 25 6 15 12 8 34 9 18 2	K1
3.	Show the bubble sort results for each pass for the following initial array of elements 35 18 7 12 5 23 16 3 1	K2
4.	Illustrate with example the open addressing and chaining methods of techniques collision resolution techniques in hashing.	K2
5.	i) Write and explain linear search procedure or algorithm with a suitable example. (8) ii) Briefly explain binary search. Search the number 26 from the list of numbers and give the steps. 10,7,17,26,32,92 (8)	K3
6.	Given input {4371, 1323, 6173, 4199, 4344, 9679, 1989} and a hash function $h(x) = x \bmod 10$. Prepare the result for the following: i) Separate chaining hash table. (4) ii) Open addressing hash table using linear probing. (4) iii) Open addressing hash table using quadratic probing. (4) iv) Open addressing hash table with second hash $h_2(x) = 7 - (x \bmod 7)$. (4)	K6